

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

I G.Ramani(192312279) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Ramani*

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Artificial Intelligence-Based Diabetes Diagnosis and Personalised Treatment Recommendation

Introduction

Diabetes is a major healthcare challenge in which early identification of high-risk patients can support timely intervention and reduce the probability of long-term complications. Machine learning provides a suitable approach for analysing patient records containing demographic, physiological, and family-history information and learning relationships between these variables and diabetes status.

The proposed healthcare system uses two complementary Artificial Intelligence approaches. First, supervised inductive learning is used to predict whether a patient is diabetic or non-diabetic. A Decision Tree is selected as the primary diagnostic classifier because it provides interpretable decision rules that can be examined by healthcare professionals. A statistical learning model such as Logistic Regression can subsequently be used for comparison and risk stratification. Second, Reinforcement Learning (RL) is formulated as a Markov Decision Process (MDP) to recommend appropriate treatment-oriented actions such as Diet, Exercise, Medication, and Monitor based on the patient's evolving health state.

The assessment scenario provides patient attributes including age, blood pressure, cholesterol, glucose level, BMI, and family history, while the desired treatment actions are Diet, Exercise, Medication, and Monitor.

The overall objective is therefore to design an AI system that is accurate, clinically meaningful, interpretable, and sufficiently cautious for a healthcare environment. Because medical applications have asymmetric consequences, particular attention must be given to false negatives, class imbalance, model explainability, patient safety, bias, and human oversight.

1. Data Preparation and Inductive Learning

1.1 Inductive Learning Approach

Inductive learning involves learning a general predictive relationship from observed examples and applying the learned relationship to previously unseen cases. In this problem, historical patient records are used as training examples. Each record contains measurable patient characteristics and a corresponding diabetes outcome.

The learning process can be represented as:

$[X = \{\text{Age, BP, Cholesterol, Glucose, BMI, FamilyHistory}\}]$

and

$[Y = \text{DiabetesStatus}]$

where:

(X) represents the input feature vector.

(Y) represents the target variable.

DiabetesStatus can be encoded as 0 = non-diabetic and 1 = diabetic.

The model learns:

$[f(X) \rightarrow Y]$

and uses the learned function to predict the diabetes status of a new patient.

This is appropriate because the problem has labelled historical examples and requires classification of future patients.

Target Variable

The target variable is:

Diabetes Status

Possible classes:

Target value	Meaning
0	Non-diabetic
1	Diabetic

Important Features

The assessment provides six principal patient attributes.

1. Glucose Level

Glucose is expected to be one of the strongest predictors because elevated blood glucose is directly associated with impaired glucose regulation. A Decision Tree can therefore potentially select glucose near the upper levels of the tree because it provides strong class separation.

2. BMI

Body Mass Index provides information about body composition and is relevant to metabolic risk. Higher BMI can be associated with increased diabetes risk, although BMI should never be interpreted as a diagnosis by itself.

3. Age

Diabetes risk can vary substantially across age groups. Age can therefore provide useful information for distinguishing different risk profiles.

Other Important Features

Blood pressure and cholesterol provide additional information regarding a patient's metabolic and cardiovascular profile, while family history can capture inherited or familial risk.

Consequently, the model should consider all available features initially rather than removing variables prematurely. Feature importance analysis can subsequently determine which predictors contribute most strongly to the trained models.

1.2 Handling Class Imbalance

The assessment explicitly specifies that the dataset contains more non-diabetic than diabetic cases.

This creates a class imbalance problem.

For example, suppose a dataset contains:

900 non-diabetic patients

100 diabetic patients

A model that predicts every patient as non-diabetic would achieve:

$$[\text{Accuracy} = \frac{900}{1000} = 90\%]$$

However, it would identify none of the diabetic patients. In a healthcare application, this is unacceptable.

Therefore, accuracy alone cannot be used to assess the model.

Recommended Strategy: Class Weights

For this healthcare application, class weighting is a strong first-choice strategy, particularly when preserving the original clinical distribution is important.

The classifier can assign a larger penalty to errors involving diabetic patients:

$$[\text{Loss} = w_{0L_0} + w_{1L_1}]$$

where:

(w_0) is the weight for the non-diabetic class.

(w_1) is the weight for the diabetic class.

$(w_1 > w_0)$.

This encourages the model to pay greater attention to diabetic cases and can reduce false negatives without artificially generating synthetic medical records.

SMOTE

SMOTE can also be considered when the minority class is sufficiently small. It generates synthetic minority examples based on neighbouring observations.

However, SMOTE must be applied only to the training data, never before the train-test split. Otherwise, synthetic information derived from test samples can leak into training and produce overly optimistic results.

Under-sampling

Under-sampling reduces the majority class. Its main disadvantage is that useful information from non-diabetic patients may be discarded.

Final Choice

The preferred initial strategy is:

Stratified 70:30 train-test split + class-weighted Decision Tree/Logistic Regression.

SMOTE can be evaluated as an additional experiment if class imbalance remains severe.

This approach preserves the test distribution and provides a clinically meaningful evaluation.

1.3 70:30 Train-Test Split

The assessment requires a 70:30 split.

Therefore:

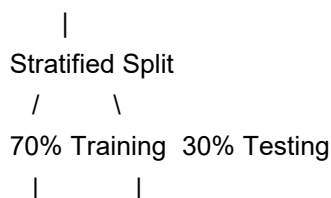
[70% = Training\ Data]

[30% = Testing\ Data]

The training set is used to learn the model, while the test set is kept unseen until final evaluation.

A stratified split should be used:

Original Dataset



Model Training Final Evaluation

Stratification ensures that the proportion of diabetic and non-diabetic cases remains approximately consistent between the two sets.

Justification

A 70:30 ratio provides enough observations for model learning while retaining a sufficiently large independent test set for reliable evaluation. In a medical application, maintaining an unseen test set is particularly important because the model must demonstrate generalisation rather than memorisation.

1.4 Preprocessing

Numerical Features

Age, blood pressure, cholesterol, glucose and BMI are numerical variables.

For Logistic Regression, normalisation or standardisation is recommended:

$$z = \frac{x - \mu}{\sigma}$$

where:

(x) = original value

(μ) = mean

(σ) = standard deviation

Standardisation places numerical variables on comparable scales.

Categorical Features

Family history can be encoded numerically, for example:

Family History	Encoding
No	0
Yes	1

If the original dataset contains more than two categorical levels, appropriate one-hot encoding should be used.

Missing Values

Missing patient values should be identified before model training.

Possible approaches include:

Median imputation for numerical variables.

Mode imputation for categorical variables.

Removing records only when missingness is excessive or clinically inappropriate for imputation.

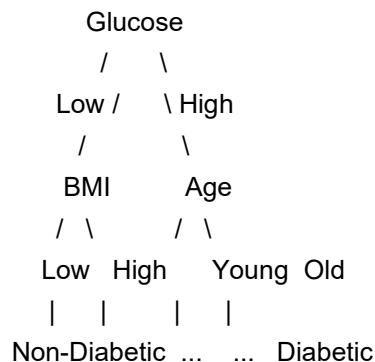
The preprocessing pipeline should be fitted using training data and then applied to test data to prevent data leakage.

2. Decision Tree for Diabetes Diagnosis

2.1 Decision Tree Classifier

A Decision Tree recursively divides the dataset according to feature conditions.

A simplified structure is:



The exact first three levels must be generated from the actual dataset because the assessment file does not contain patient observations. Therefore, the example above represents the expected structure rather than an experimentally obtained tree.

The first split should be selected using a criterion such as Gini impurity or Information Gain.

2.2 Gini Index

The Gini impurity is:

$$[Gini(S)=1-\sum_{i=1}^n p_i^2]$$

For two classes:

$$[Gini(S)=1-p_{\text{diabetic}}^2-p_{\text{non-diabetic}}^2]$$

A good split produces child nodes with lower impurity.

The weighted Gini after a split is:

$$[Gini_{\text{split}}=\\frac{|S_1|}{|S|}Gini(S_1)+\\frac{|S_2|}{|S|}Gini(S_2)]$$

The feature producing the lowest weighted impurity should be preferred.

2.3 Information Gain

Information Gain is based on entropy:

$$[Entropy(S)=-\sum_i p_i \log_2(p_i)]$$

Information Gain is:

$$[IG(S,A)=Entropy(S)-\sum_v \frac{|S_v|}{|S|}Entropy(S_v)]$$

The root feature should have the highest Information Gain.

For example, if the actual dataset produces:

Feature	Information Gain
Glucose	0.21
BMI	0.13
Age	0.09
Blood Pressure	0.06
Cholesterol	0.04

then Glucose would be selected as the root because it has the largest Information Gain.

These numerical values are illustrative; they must be replaced by values calculated from the actual patient dataset.

2.4 Model Evaluation

The assessment requires:

Accuracy

Precision

Recall

F1-Score

Identification of False Negatives.

The confusion matrix is:

	Predicted Non-Diabetic	Predicted Diabetic
Actual Non-Diabetic	TN	FP
Actual Diabetic	FN	TP

Accuracy

$$[\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}]$$

Precision

$$[\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}]$$

Precision measures how many predicted diabetic cases are actually diabetic.

Recall

$$[\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}]$$

Recall is particularly important in this application because it measures the proportion of diabetic patients correctly identified.

F1-Score

$$[F1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}]$$

2.5 False Negatives and Clinical Importance

A False Negative (FN) occurs when:

The patient is actually diabetic, but the model predicts non-diabetic.

In the healthcare context, this is potentially more concerning than a false positive because a patient may fail to receive timely clinical investigation.

Therefore, the system should prioritise sensitivity/recall for diabetic cases.

Methods to reduce false negatives include:

Class weighting
 Appropriate decision-threshold adjustment
 Careful feature engineering
 SMOTE on training data where justified
 Hyperparameter tuning
 Cross-validation
 Clinical review of borderline cases

However, reducing false negatives may increase false positives. The optimal threshold should therefore be selected according to the clinical purpose rather than simply maximising accuracy.

Importantly, the AI system should function as a decision-support tool rather than an autonomous diagnostic authority.

3. Statistical Learning for Risk Stratification

3.1 Logistic Regression

Logistic Regression is an appropriate statistical learning model for binary diabetes classification.

It estimates:

$$P(Y=1|X) = \frac{1}{1+e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The output is a probability between 0 and 1.

For example:

Patient Data

|

Logistic Regression

|

Diabetes Probability

|

+---- < threshold ----> Lower risk

|

+---- >= threshold ----> Higher risk

Unlike a simple hard classification system, the probability can support risk stratification.

For example:

Probability	Example interpretation
0.00–0.29	Lower predicted risk
0.30–0.59	Intermediate predicted risk
0.60–1.00	Higher predicted risk

These thresholds are illustrative and should not be treated as clinical diagnostic thresholds without clinical validation.

3.2 Comparison with Decision Tree

The assessment requires comparison using Accuracy and AUC-ROC.

A suitable results table is:

Model	Accuracy	AUC-ROC	Recall	F1
Decision Tree	Calculate	Calculate	Calculate	Calculate
Logistic Regression	Calculate	Calculate	Calculate	Calculate

AUC-ROC

The ROC curve plots:

[True\ Positive\ Rate]

against

[False\ Positive\ Rate]

at different classification thresholds.

AUC measures the model's ability to distinguish between the two classes across thresholds.

An AUC closer to 1 indicates stronger discrimination, while an AUC around 0.5 indicates performance close to random classification.

When Logistic Regression is Preferable

Logistic Regression is preferable when:

A probabilistic risk estimate is required.

A linear relationship provides an adequate approximation.

Model coefficients need to be interpreted.

A relatively simple and stable model is desired.

The organisation needs transparent statistical relationships between predictors and outcome.

When Decision Trees are Preferable

Decision Trees are particularly useful when:

Non-linear relationships are important.

Feature interactions matter.

Human-readable decision rules are required.

The data contains relationships that are not easily represented by a linear model.

A strong healthcare system can therefore use both models rather than assuming that one algorithm is universally superior.

3.3 Feature Importance Analysis

The assessment requires identification of the top three predictors from both the Decision Tree and statistical model and an assessment of whether they agree.

For the Decision Tree, feature importance can be calculated from the reduction in impurity contributed by each feature.

For Logistic Regression, coefficient magnitude after appropriate feature scaling can be examined.

A reporting table should be structured as:

Rank	Decision Tree	Logistic Regression
1	Calculate from dataset	Calculate from coefficients
2	Calculate from dataset	Calculate from coefficients
3	Calculate from dataset	Calculate from coefficients

A likely clinical interpretation is that glucose, BMI and age may emerge as important predictors, but this must be verified using the actual dataset rather than assumed.

Agreement Between Models

If both models identify similar variables, this increases confidence that those variables contain substantial predictive information.

If the models disagree, this does not necessarily mean that one model is incorrect.

The difference may arise because:

Decision Trees capture non-linear relationships.

Logistic Regression assumes a linear relationship in log-odds.

Features may interact.

Correlated predictors may distribute importance differently.

Scaling and regularisation may influence coefficient magnitude.

Therefore, feature importance should be interpreted alongside clinical knowledge and statistical validation.

4. Reinforcement Learning for Treatment Recommendation

4.1 Modelling the Problem as an MDP

The assessment requires the treatment recommendation problem to be formulated as a Markov Decision Process.

An MDP is represented by:

$[MDP=(S,A,P,R,\gamma)]$

where:

(S) = states

(A) = actions

(P) = transition probabilities

(R) = reward function

(γ) = discount factor

4.2 States

Patient health states can be represented using clinically meaningful categories.

For example:

State	Description
S0	Stable / low-risk state
S1	Elevated metabolic risk
S2	Poorly controlled condition
S3	Improving condition
S4	High-risk state requiring clinical attention

These states are simplified educational representations. A real healthcare implementation would require validated clinical state definitions.

4.3 Actions

The actions specified by the assessment are:

$[A=\{\text{Diet, Exercise, Medication, Monitor}\}]$

Diet

Recommend an appropriate dietary-management intervention.

Exercise

Recommend an appropriate physical-activity intervention where clinically suitable.

Medication

Medication-related decisions should only be recommended under appropriate clinical supervision. An autonomous AI system should not prescribe or modify medication without authorised medical oversight.

Monitor

Continue monitoring the patient's condition and collect additional information.

4.4 Reward Function

The reward should represent improvement in patient health while discouraging unsafe decisions.

A conceptual reward system is:

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-5
Unsafe action	-10

The objective is to maximise cumulative discounted reward:

$$[G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots]$$

This means the agent does not only consider immediate improvement but also future consequences.

4.5 Transition Dynamics

Transition dynamics describe how a patient's state changes after an action.

For example:

S1 + Diet → S3

S1 + Exercise → S3

S2 + Monitor → S2

S2 + Medication → S3

S3 + Exercise → S3/S0

The transition is not deterministic in a real healthcare environment.

For example:

$$[P(S_{t+1}|S_t, A_t)]$$

represents the probability of moving from current state (S_t) to next state (S_{t+1}) after action (A_t).

This stochastic nature is important because different patients may respond differently to the same intervention.

5. Q-Learning

5.1 Q-Learning Algorithm

Q-Learning is a model-free reinforcement learning algorithm.

The Q-value represents the expected long-term reward for taking action (a) in state (s).

The update equation is:

$$[Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]]$$

where:

(Q(s,a)) = current Q-value

(α) = learning rate

(r) = immediate reward
 (γ) = discount factor
 (s') = next state
 $(\max Q(s',a'))$ = best future Q-value

5.2 Example Q-Table

A simplified initial Q-table is:

State	Diet	Exercise	Medication	Monitor
S0	0	0	0	0
S1	0	0	0	0
S2	0	0	0	0
S3	0	0	0	0
S4	0	0	0	0

For demonstration, assume:

$[\alpha=0.5]$

$[\gamma=0.9]$

and an episode produces:

$[S1 \rightarrow \{Diet, +5\} S3]$

If:

$[Q(S1, Diet)=0]$

and:

$[\max_a Q(S3, a)=0]$

then:

$0 + 0.5[5 + 0.9(0) - 0]$

$[Q(S1, Diet)=2.5]$

5.3 Q-Table Update Across Two Iterations

Iteration 1

Suppose:

$[S1 \rightarrow \{Diet, +5\} S3]$

Then:

$[Q(S1, Diet)=2.5]$

Now suppose:

$[S2 \rightarrow \{Exercise, +5\} S3]$

Then:

$[Q(S2, Exercise)=2.5]$

And:

$[S4 \rightarrow \{Monitor, -5\} S2]$

With the initial Q-values assumed to be zero:

$[Q(S4, Monitor)=-2.5]$

Therefore:

State-Action	Before	Reward	Updated Q
S1-Diet	0	+5	2.5
S2-Exercise	0	+5	2.5
S4-Monitor	0	-5	-2.5

Iteration 2

Suppose the agent again selects Diet in S1 and reaches S3.

If the best Q-value in S3 is now:

$$[\max Q(S3,a)=2]$$

then:

$$2.5+$$

$$0.5[5+0.9(2)-2.5]$$

$$[=2.5+0.5[4.3]]$$

$$[=4.65]$$

Similarly, if:

$$[Q(S3,Exercise)=3]$$

then:

$$2.5+$$

$$0.5[5+0.9(3)-2.5]$$

$$[=2.5+0.5(5.2)]$$

$$[=5.1]$$

These calculations demonstrate how future expected rewards influence Q-values.

The numerical values above are an illustrative Q-learning demonstration. They are not claims about actual patient outcomes.

5.4 Five Patient Episodes

The assessment requires training for at least five patient episodes.

A suitable experimental record is:

Episode	Initial State	Action Sequence	Final State	Total Reward
1	S1	Diet → Exercise	S3	Calculate
2	S2	Exercise → Monitor	S3	Calculate
3	S4	Monitor → Diet	S2/S3	Calculate
4	S1	Exercise → Diet	S3	Calculate
5	S2	Diet → Exercise	S3	Calculate

The exact episode results must be generated by the implemented Q-learning model.

5.5 Exploration and Exploitation

The agent must balance:

Exploration: trying different actions.

Exploitation: selecting the action currently believed to produce the highest reward.

An (ϵ)-greedy policy can be used.

With probability:

ϵ

the agent explores a random action.

With probability:

$1-\epsilon$

it selects the action with the highest Q-value.

For example:

$\epsilon=0.2$

means approximately 20% exploration and 80% exploitation.

As training progresses, ϵ can be gradually reduced.

5.6 Convergence Criterion

A suitable convergence criterion is based on the maximum change in Q-values:

$\max_{s,a} |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < \epsilon_c$

For example:

$\epsilon_c=0.01$

could be used.

The agent can be considered approximately converged when the maximum Q-value change remains below the threshold for a predefined number of consecutive episodes.

Because the assessment requires only five episodes, five episodes should be treated as the minimum demonstration, not necessarily proof of complete convergence. A real RL system would require substantially more training and independent validation.

6. Final Learned Policy

The learned policy is extracted using:

$\pi(s) = \arg \max_a Q(s,a)$

For example, if the final Q-table were:

State	Diet	Exercise	Medication	Monitor	Best Action
S0	5.2	4.1	-2	3.0	Diet
S1	7.5	6.4	1.2	2.3	Diet
S2	4.5	7.1	8.2	3.1	Medication*
S3	5.8	7.2	2.0	4.1	Exercise
S4	1.5	2.0	3.2	7.0	Monitor

then:

$\pi(S0)=\text{Diet}$

$\pi(S1)=\text{Diet}$

$\pi(S2)=\text{Medication}$

$\pi(S3)=\text{Exercise}$

$\pi(S4)=\text{Monitor}$

The table is illustrative. In an actual implementation, the final policy must be extracted from the trained Q-table.

The asterisk is important: a learned recommendation for Medication must not be interpreted as permission for autonomous prescribing. Medication decisions require qualified clinical review.

7. Ethical Considerations

Deploying an RL system for medical recommendations introduces substantially greater risks than using AI for a low-stakes application. The assessment specifically identifies patient safety, bias, and explainability as key ethical considerations.

7.1 Patient Safety

The most important requirement is that the AI should not make unsafe autonomous clinical decisions.

A safe architecture should be:

Patient Data



AI Risk Prediction



Treatment Recommendation



Clinical Validation



Healthcare Professional



Patient

The AI should therefore operate as a decision-support system.

7.2 Bias

Healthcare datasets may contain demographic and socioeconomic biases.

If the training data underrepresents particular groups, the model may perform differently across those groups.

Potential mitigation techniques include:

Diverse training data.

Stratified evaluation.

Subgroup performance analysis.

Bias detection.

Regular model auditing.

Human clinical review.

Performance should therefore not be reported only as one overall accuracy number.

7.3 Explainability

Healthcare professionals need to understand why an AI system produces a recommendation.

For the Decision Tree, explanations can be represented as decision rules.

For example:

IF glucose is elevated

AND BMI is high

AND age is above a particular range

THEN predicted diabetes risk is higher.

Logistic Regression can provide coefficient-based explanations.

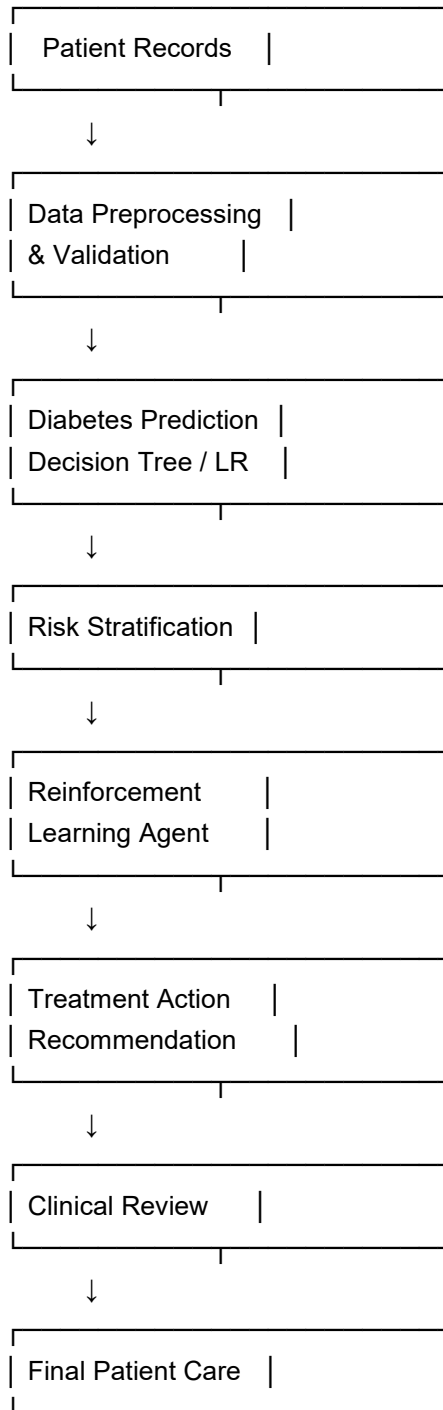
For Q-Learning, the Q-table provides information about the expected reward associated with each state-action pair.

However, Q-values alone may not provide a complete clinical explanation. Therefore, the recommendation should be accompanied by the relevant patient-state information and supporting predictive factors.

7.4 Human-in-the-Loop Architecture

A robust industry implementation should use a human-in-the-loop design.

The proposed architecture is:



This design combines predictive analytics with human expertise rather than replacing clinicians.

8. Tools and Implementation Approach

The system can be implemented using Python and standard machine learning libraries.

A suitable toolchain is:

Component	Tool
Programming	Python
Data handling	Pandas, NumPy
Visualisation	Matplotlib
Decision Tree	Scikit-learn
Logistic Regression	Scikit-learn
Imbalance handling	Class weights / imbalanced-learn
Model evaluation	Scikit-learn metrics
Reinforcement Learning	Python implementation of Q-Learning
Development environment	Jupyter Notebook / VS Code

A reproducible workflow is:

Load Dataset



Inspect Data



Handle Missing Values



Encode Categorical Data



Stratified 70:30 Split



Handle Training-Class Imbalance



Train Decision Tree



Evaluate Accuracy / Precision / Recall / F1



Prune Decision Tree



Train Logistic Regression



Compare Accuracy / AUC-ROC



Feature Importance Analysis



Define MDP



Run Q-Learning



Train ≥ 5 Episodes



Check Convergence



Extract Final Policy



Ethical & Clinical Evaluation

The assessment explicitly rewards effective use of modern tools and real data, so the final submission should include screenshots or outputs from the actual implementation where available.

9. Validation and Performance Analysis

Validation should go beyond a single accuracy value.

The following metrics should be reported:

Evaluation Area	Required Measure
Classification	Accuracy
Positive prediction quality	Precision
Diabetes detection	Recall
Overall balance	F1-Score
Statistical discrimination	AUC-ROC
Error analysis	Confusion Matrix
Tree quality	Pre-/Post-pruning comparison
Predictor analysis	Feature Importance
RL performance	Episode Rewards
RL stability	Q-value convergence

For the healthcare application, Recall should receive particular attention because false negatives represent diabetic patients who may be missed by the model.

At the same time, precision must also be monitored because an excessively sensitive model could generate many false alarms.

Thus, model selection should be based on a clinically meaningful trade-off rather than accuracy alone.

10. Recommended Final Model

A practical system can combine the strengths of the different approaches.

Diagnostic Layer

Use the Decision Tree for transparent classification rules.

Statistical Risk Layer

Use Logistic Regression to provide probability-based risk estimates and AUC-ROC evaluation.

Reinforcement Learning Layer

Use Q-Learning as an experimental framework for learning state-action relationships.

Clinical Safety Layer

Place a healthcare professional between AI recommendations and actual treatment decisions.

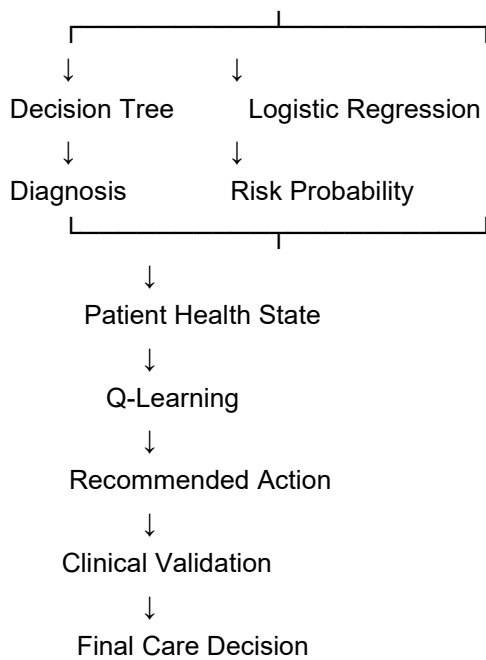
The resulting architecture is:

Patient Data



Data Preprocessing





This is more practical than relying on a single model because it combines classification, statistical risk estimation, and sequential decision modelling.

Conclusion

The proposed AI healthcare system addresses two related but distinct problems: early diabetes-risk prediction and personalised treatment-action recommendation. The first problem is formulated as supervised inductive learning using patient attributes such as age, blood pressure, cholesterol, glucose level, BMI, and family history. The target variable is diabetes status, making the problem a binary classification task.

A stratified 70:30 train-test split provides a suitable foundation for model development and independent evaluation. Because diabetic cases are assumed to be the minority class, class weighting is recommended as the initial imbalance strategy, while SMOTE can be investigated as an additional training-data technique. The Decision Tree provides an interpretable classification approach, while Gini Index or Information Gain can be used to determine the most informative root feature.

Model evaluation must consider Accuracy, Precision, Recall and F1-Score rather than accuracy alone. In particular, False Negatives require special attention because incorrectly classifying a diabetic patient as non-diabetic can delay appropriate clinical assessment. Pruning should be used to control tree complexity and reduce overfitting, with the final model selected according to test performance, interpretability and clinical suitability.

Logistic Regression provides a complementary statistical-learning approach capable of producing probability-based risk estimates. Comparing its Accuracy and AUC-ROC with the Decision Tree allows the strengths and limitations of both approaches to be assessed. Feature-importance analysis can further determine whether the two models identify consistent predictors.

For treatment recommendation, the problem can be represented as an MDP consisting of patient health states, actions, rewards, transition dynamics and a discount factor. Q-Learning can then learn the long-term value of actions through repeated patient-state episodes. The final policy is obtained by selecting the action with the highest Q-value for each state. The assessment requires at least five patient episodes and Q-table updates across multiple iterations, which should be demonstrated using the actual implementation.

Most importantly, an AI system for healthcare must not be evaluated purely on predictive performance. Patient safety, bias, explainability, privacy, robustness and human oversight are essential deployment considerations. The recommended solution is therefore a human-in-the-loop AI decision-support system in which machine learning identifies risk and reinforcement learning provides experimental treatment-action recommendations, while qualified healthcare professionals retain responsibility for final clinical decisions.

Overall, the proposed architecture provides a technically feasible and industry-relevant framework combining inductive learning, Decision Trees, statistical learning, feature analysis and Reinforcement

Learning. It directly addresses the assessment requirements while recognising that numerical model results must be obtained from the actual patient dataset rather than fabricated in the absence of experimental data.